

Powers and Modular Exponentiation

Q&A Notes

Question 1

Draw a table showing the order of every element in $\mathbb{Z}_{11}^*$. Pick the smallest primitive root g and compute a table giving $\text{ind}_{11,g}(x)$ for all $x \in \mathbb{Z}_{11}^*$.

Answer 1

The smallest primitive root modulo 11 is

$$g = 2.$$

The group $\mathbb{Z}_{11}^*$ contains the nonzero residues modulo 11:

$$\mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}.$$

The order of an element a is the smallest positive integer t such that

$$a^t \equiv 1 \pmod{11}.$$

Element	Order	Reason
1	1	$1^1 \equiv 1 \pmod{11}$.
2	10	Powers are 2, 4, 8, 5, 10, 9, 7, 3, 6, 1.
3	5	Powers are 3, 9, 5, 4, 1.
4	5	Powers are 4, 5, 9, 3, 1.
5	5	Powers are 5, 3, 4, 9, 1.
6	10	Powers are 6, 3, 7, 9, 10, 5, 8, 4, 2, 1.
7	10	Powers are 7, 5, 2, 3, 10, 4, 6, 9, 8, 1.
8	10	Powers are 8, 9, 6, 4, 10, 3, 2, 5, 7, 1.
9	5	Powers are 9, 4, 3, 5, 1.
10	2	$10^2 = 100 \equiv 1 \pmod{11}$.

A primitive root modulo 11 is an element with order

$$\phi(11) = 10.$$

The smallest element with order 10 is 2, so

$$g = 2.$$

Now compute the index table. The value

$$\text{ind}_{11,2}(x)$$

means the exponent k such that

$$2^k \equiv x \pmod{11}.$$

k	0	1	2	3	4	5	6	7	8	9
$2^k \pmod{11}$	1	2	4	8	5	10	9	7	3	6

So the index table is:

x	$\text{ind}_{11,2}(x)$
1	0
2	1
3	8
4	2
5	4
6	9
7	7
8	3
9	6
10	5

Answer Summary

$$g = 2$$

and

$$\text{ind}_{11,2}(1, 2, 3, 4, 5, 6, 7, 8, 9, 10) = (0, 1, 8, 2, 4, 9, 7, 3, 6, 5).$$

Question 2

Give a modular exponentiation algorithm that examines the bits of b from right to left instead of left to right.

Answer 2

A right-to-left modular exponentiation algorithm keeps two values:

Variable	Meaning
d	Accumulates the answer.
$base$	Stores the current power of a , repeatedly squared.

Right-to-Left Modular Exponentiation

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RIGHT-TO-LEFT-MODULAR-EXPONENTIATION(a, b, n):
    d = 1
    base = a mod n

    while b > 0:
        if b is odd:
            d = (d * base) mod n

        base = (base * base) mod n
        b = floor(b / 2)

    return d

```

Why this works:

The binary expansion of b is

$$b = b_0 + 2b_1 + 4b_2 + \cdots + 2^k b_k,$$

where each bit b_i is 0 or 1.

So

$$a^b = a^{b_0} \cdot a^{2b_1} \cdot a^{4b_2} \cdots a^{2^k b_k}.$$

The algorithm scans the bits from right to left:

- If the current bit is 1, multiply the answer by the current base.
- Square the base each time, so it becomes $a, a^2, a^4, a^8, \dots$
- Divide b by 2 to move to the next bit.

Key Difference

Left-to-right exponentiation reads the binary representation from the most significant bit.

Right-to-left exponentiation reads it from the least significant bit using repeated division by 2.

Question 3

Assuming that you know $\phi(n)$, explain how to compute

$$a^{-1} \bmod n$$

for any $a \in \mathbb{Z}_n^*$ using the procedure MODULAR-EXPONENTIATION.

Answer 3

The inverse is

$$a^{-1} \equiv a^{\phi(n)-1} \pmod{n}.$$

Since

$$a \in \mathbb{Z}_n^*,$$

we know

$$\gcd(a, n) = 1.$$

By Euler's theorem,

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Rewrite the left side:

$$a^{\phi(n)} = a \cdot a^{\phi(n)-1}.$$

So

$$a \cdot a^{\phi(n)-1} \equiv 1 \pmod{n}.$$

This means $a^{\phi(n)-1}$ is the multiplicative inverse of a modulo n :

$$a^{-1} \equiv a^{\phi(n)-1} \pmod{n}.$$

Therefore, to compute the inverse, run:

Inverse by Modular Exponentiation

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MODULAR-INVERSE-BY-EULER(a, n):  
    return MODULAR-EXPONENTIATION(a, phi(n) - 1, n)
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Example

Find

$$5^{-1} \bmod 12.$$

First compute

$$\phi(12) = 4.$$

Then

$$5^{-1} \equiv 5^{4-1} \equiv 5^3 \pmod{12}.$$

Compute:

$$5^2 = 25 \equiv 1 \pmod{12}.$$

So

$$5^3 \equiv 5 \pmod{12}.$$

Therefore

$$\boxed{5^{-1} \equiv 5 \pmod{12}.$$

Check:

$$5 \cdot 5 = 25 \equiv 1 \pmod{12}.$$